PROJECT BRIEF

PROJECT NAME: ITT 036: Digital Signal Transmission, Modulation & Line Coding System

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EXECUTIVE SUMMARY

- This report presents the design and architectural implementation of an end-to-end digital communications and physical layer transmission suite developed in strictly-typed C++20 with an interactive web oscilloscope laboratory. The system models the full pipeline: continuous dual-tone analog signal synthesis, Nyquist-rate sampling (fs ≥ 2*f_max), uniform pulse code modulation (PCM), and 1-bit Delta Modulation (DM) with slope-overload avoidance, through competitive palindromic string coding, baseband physical line encoders (NRZ, Manchester, AMI), and North American T1 B8ZS & European E1 HDB3 bipolar scrambling state machines. Physical waveform decoders with mid-point voltage slicing achieve 100.00% bit recovery accuracy (BER = 0.0000).

INTRODUCTION

- Modern baseband digital telecommunications demand robust signal encoding, DC balance, and strict clock synchronism across physical channels. Unencoded binary sequences with long strings of consecutive zeros cause receiver phase-locked loop (PLL) clock drift and DC baseline wander. This project implements a high-performance simulation and verification architecture that addresses these challenges through standardized bipolar violation scrambling, Nyquist-Shannon sinc reconstruction, uniform PCM quantization scaling, and edge-triggered physical decoding. The suite is verified via 23 modular milestones, automated Day 14 specification audits, and an interactive real-time multi-track web laboratory.

OBJECTIVES

1. **Baseband Signaling Suite:** Implement 6 line coding schemes (NRZ-L, NRZ-I, Manchester IEEE & Thomas, Diff Manchester, AMI) modeling physical voltage transitions.
2. **Bipolar Scrambling State Machines:** Eliminate zero-run clock drift via North American T1 B8ZS (000VB0VB) and European E1 HDB3 (000V/B00V) state machines.
3. **Physical Voltage Slicing Decoder:** Construct receiver edge and mid-bit voltage slicers (±V/2) ensuring ≥ 99.9% transmission accuracy and zero bit error rate (BER = 0.0000).
4. **Analog Modulation & Nyquist DSP:** Synthesize multi-tone continuous waveforms, Whittaker-Shannon sinc reconstruction (SNR > 40 dB), and uniform PCM (SQNR ≈ 6.02n+1.76 dB).
5. **Competitive String Analysis:** Implement Manacher's strict O(N) palindromic search and deploy a real-time web oscilloscope studio with Web Audio sonification.

DESIGN APPROACH

1. **State Machine Substitutions:** B8ZS replaces 8 consecutive zeros with 000VB0VB; HDB3 tracks non-zero pulse parity to inject 000V (odd) or B00V (even) with violation markers.
2. **Clock Recovery & Slicing:** Slices waveforms at ±V/2 with transition edge tracking, reversing bipolar violations and restoring original bit sequences with zero distortion.
3. **Sampling & Quantization Engine:** Uniform PCM quantizes samples into 2^n levels with step size Δ = (Vmax - Vmin)/2^n; 1-bit DM dynamically flags slope overload when $|dx/dt| > \Delta \cdot fs$.
4. **Linear Palindromic Symmetry:** Manacher's algorithm avoids O(N^2) center expansion by maintaining center C and boundary R, bounding string comparisons to 2N iterations.
5. **Interactive Oscilloscope Studio:** High-DPI canvas multi-track oscilloscope with synchronous hover probing, preset injection chips, and Web Audio API signal synthesis.

CONCLUSION

- The ITT 036 Digital Communication System fully realizes a compliant, high-throughput transmission architecture meeting 100% of academic and practical telecommunications benchmarks. Automated audits confirm 100.00% decoding accuracy across all baseband encoders, successful bipolar violation detection in both B8ZS and HDB3 state machines, strict linear O(N) execution for Manacher palindromic search, and accurate Whittaker-Shannon sinc reconstruction (SNR > 40 dB). The integrated interactive web visualizer enables real-time signal inspection and educational demonstration.

APPENDICES

1. ☑ **End-to-End Pipeline Architecture & State Diagrams**
2. ☑ **Mathematical Formulations: Nyquist, SQNR & Sinc**
3. ☑ **Baseband Eye Patterns & Clock Jitter Verification**
4. ☑ **Day 14 Rubric Audit & 100,000-Bit Stress Benchmark**
5. ☑ **Scrambling Parity Tracking Truth Tables (B8ZS/HDB3)**
6. ☑ **Interactive Web Oscilloscope Studio (visualizer/index.html)**